CLINICAL · EAR

Tympanic membrane: trauma, retraction, barotrauma and sequelae

The drum is thin, and it records what happens around it. A traumatic hole usually heals by itself; a retracted drum is a sign of a failing Eustachian tube; and a sudden pressure injury can reach through the middle ear into the inner ear. In every case, check for vertigo, facial weakness and sensorineural hearing loss.

AT A GLANCE

- Most traumatic perforations heal on their own: keep dry.
- Vertigo or sensorineural loss needs urgent review.
- Retraction reflects tube failure.
- Barotrauma: decongestants, then myringotomy if needed.

1 CHAPTER ONE Overview

Injuries and scars of the drum.

1.1 The normal drum

A three-layered membrane: outer squamous epithelium, a middle fibrous layer, and inner mucosa. The fibrous layer gives it strength and shape; when it is lost after healing, the drum becomes thin and floppy. On otoscopy the normal right drum shows the handle of malleus, the lateral process and a bright cone of light at five o'clock.

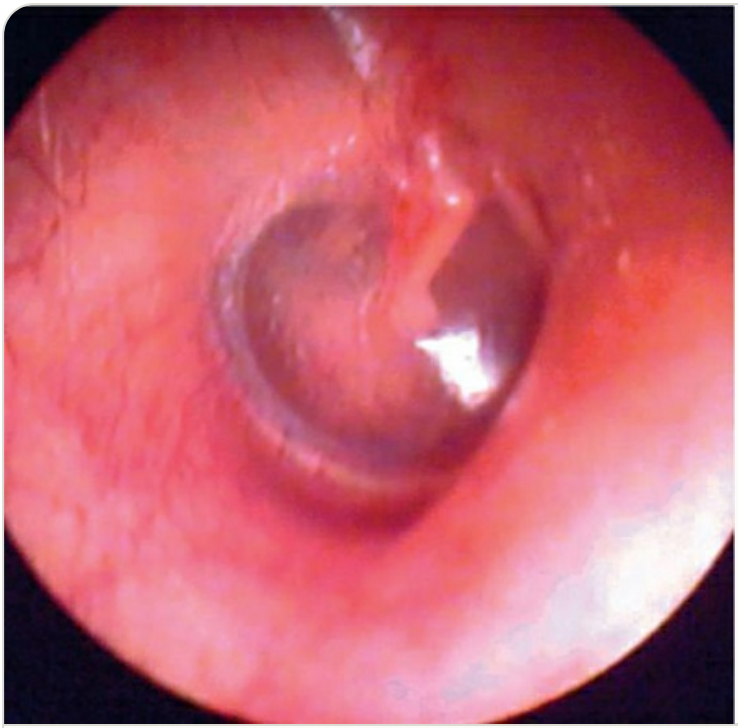


FIGURE 1 A normal right tympanic membrane, with the cone of light at five o'clock.
Dhingra, fig. 8.17

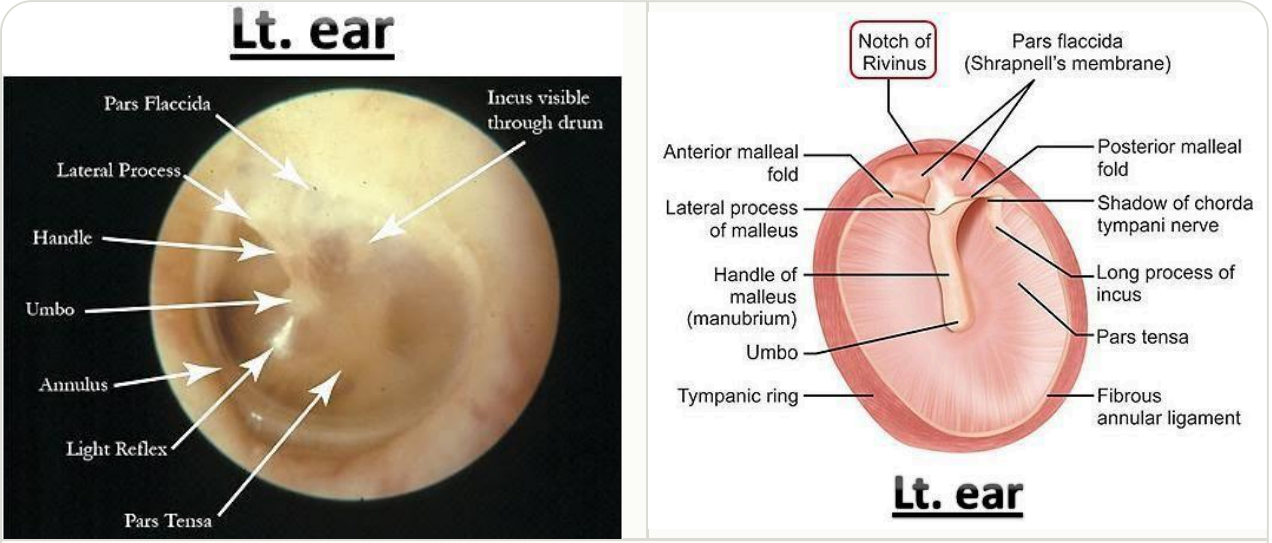


FIGURE 2 A normal tympanic membrane seen through the otoscope, and its parts.
Anatomy atlas · ENT Steps p. 36

1.2 Conditions covered

- Traumatic perforation.
- Retraction, atelectasis and retraction pockets.
- Atrophic drum and tympanosclerosis.

- Barotrauma of the middle ear.

2 CHAPTER TWO Traumatic perforation

Most heal on their own.

2.1 Causes

- **Direct:** cotton buds, hairpins, matchsticks, careless syringing or instrument use.
- **Pressure:** a slap on the ear, blast injury, a fall onto water, a kiss on the ear.
- **Temporal bone fracture.**
- **Thermal:** welding sparks or hot slag, which heal poorly.

2.2 Clinical picture

Sudden pain, a little bleeding, deafness and tinnitus. The perforation is irregular or slit-like, often with blood clot and with its edges turned inwards.



FIGURE 3 A traumatic perforation from blast, with ecchymosis (white arrow head) and tears of the drum (black arrows).

Dhingra, fig. 8.19

LOOK DEEPER

Vertigo, sensorineural hearing loss or facial weakness suggests stapes subluxation, perilymph fistula or a temporal bone fracture. Arrange urgent assessment and consider exploration.

2.3 Management

2.3.1 Assess

Tuning forks, nystagmus and the facial nerve; an audiogram when practical. CT if a fracture is suspected.

2.3.2 Keep dry, keep out

No drops, no syringing and no water in the ear. Antibiotics only if the ear becomes infected.

2.3.3 Repair

Most perforations heal within weeks. Large ones with in-turned edges may be repositioned and splinted with a paper patch under the microscope. Those that persist beyond about three months are repaired by myringoplasty.

OPERATIVE GUIDES IN THEATRE 3 operations

Myringoplasty and Tympanoplasty
Repair if it persists

Temporal Bone Trauma
Temporal bone trauma

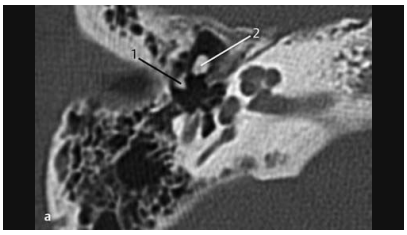
Otitis Media, Myringotomy, Tympanostomy Tube, and Balloon Dilation
Ventilation tube

2.4 Complications

Persistent perforation, infection, ossicular dislocation, perilymph fistula, and implantation cholesteatoma from squamous epithelium driven into the middle ear.

IMAGING IN THE APP 1 case

Ossicular luxation after trauma.



Luxation of the Ossicular Chain

Trauma · 10 images

3 CHAPTER THREE Retraction and atelectasis

The drum sucked in.

3.1 Why it happens

When the Eustachian tube fails, middle ear gas is absorbed and negative pressure draws the drum inwards. Healed or atrophic areas, which lack the fibrous layer, retract first.

3.2 Signs of a retracted drum

- A dull drum with a broken or absent light reflex.
- A prominent lateral process of the malleus.
- A foreshortened handle of malleus.
- Prominent anterior and posterior malleolar folds.

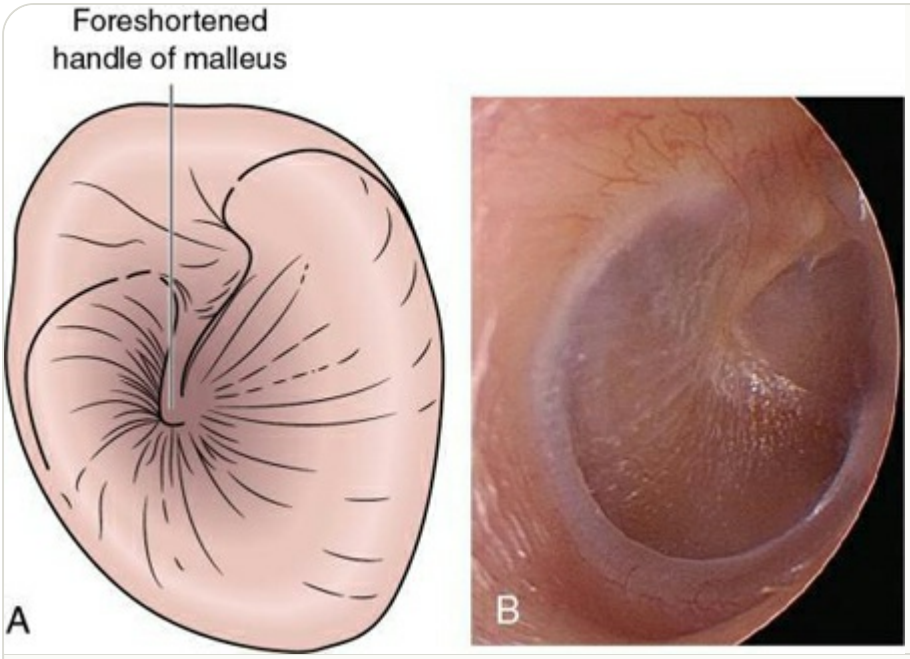


FIGURE 4 Features of a retracted tympanic membrane.
Dhingra, fig. 8.18

3.3 Retraction pockets and atelectasis

STAGES OF PARS TENSA RETRACTION

STAGE	APPEARANCE
I	Retracted, not touching the incus
II	Touching the incus
III	Lying on the promontory (atelectasis), can still be lifted
IV	Adherent to the promontory (adhesive otitis), no middle ear space

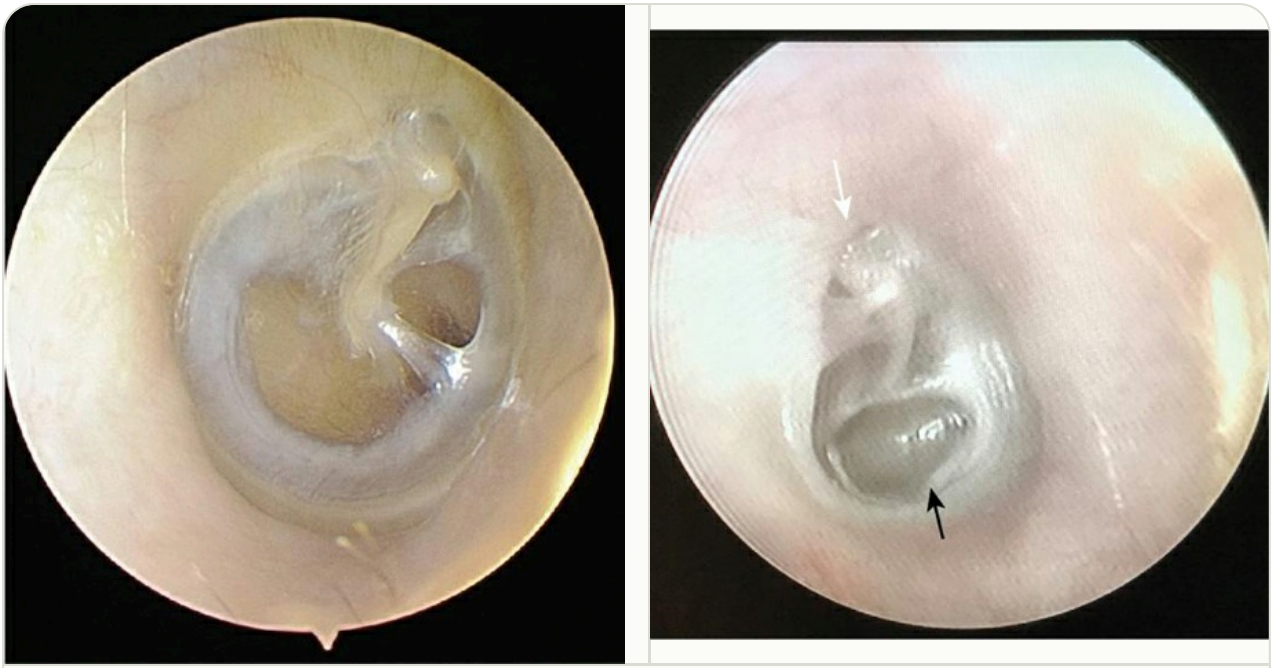


FIGURE 5 First, an atrophic drum with a retraction pocket and an adhesion from the malleus. Second, atelectasis with retraction pockets in the attic (white arrow) and the pars tensa (black arrow).

Dhingra, figs 8.20 and 8.21

3.4 Management

Treat the cause of tubal failure (adenoids, sinusitis, allergy). A grommet re-aerates the middle ear while the drum can still be lifted (stage III). Deep pockets that collect keratin, or that cannot be cleaned, are excised and reinforced with cartilage before they become cholesteatoma.

4 CHAPTER FOUR Atrophy and tympanosclerosis

Scars of past disease.

4.1 Atrophic drum

A thin, transparent segment without a fibrous layer, after healed perforations or long-standing effusion. It moves excessively and retracts easily.

4.2 Tympanosclerosis

Chalky white hyaline and calcified plaques in the fibrous layer of the drum and sometimes in the middle ear around the ossicles, after recurrent otitis media or grommets. Plaques on the drum usually cause no symptoms; deposits around the ossicles can fix them and cause a conductive loss that is difficult to correct surgically.



FIGURE 6 Tympanosclerosis with a central perforation: chalky plaques lie in front of and behind the hole.
Dhingra, fig. 8.22

5 CHAPTER FIVE Barotrauma

Aero-otitis media.

5.1 Mechanism

On descent in an aircraft or when diving, outside pressure rises. If the Eustachian tube does not open, middle ear pressure falls relative to it. Once the difference reaches about 90 mm Hg, the soft tissues at the pharyngeal end of the tube are

pushed together and the tube *locks*, so swallowing and Valsalva can no longer open it. A cold, allergy or sleeping through descent make this more likely.

5.2 Clinical picture

Severe pain, deafness, tinnitus and sometimes vertigo. The drum is retracted and congested, with haemorrhage into it, and later an effusion or blood (haemotympanum), or a perforation.

5.3 Inner ear barotrauma

Forceful Valsalva against a locked tube can rupture the round or oval window, causing sensorineural loss and vertigo: a perilymph fistula that may need exploration.

5.4 Treatment

Mild cases: decongestant drops or spray, analgesia and gentle Valsalva. Persistent effusion or failure: politzerisation or myringotomy, and a grommet for recurrent problems.

> **PREVENTION**
>
> Avoid flying with a cold, use a decongestant spray before descent, stay awake on descent, and swallow, yawn or perform Valsalva repeatedly.

6 CHAPTER SIX For the examination

What the examiner wants to hear.

6.1 Viva essentials

> **WHAT THE EXAMINER IS LOOKING FOR**
>
> - **Traumatic perforation:** causes, keep dry, most heal.

- **Red flags:** vertigo, sensorineural loss, facial weakness.
- **Signs of a retracted drum.**
- **Retraction staging,** and when a grommet helps.
- **Barotrauma:** the locked tube at about 90 mm Hg, treatment and prevention.

6.2 Common pitfalls

COMMON PITFALLS

- Syringing or putting drops into an ear with a fresh traumatic perforation.
- Missing a perilymph fistula in a patient with vertigo after a slap or a dive.
- Ignoring a deep retraction pocket until it becomes a cholesteatoma.

Written for this app from Oxford Textbook of Otolaryngology ch. 21, Oxford Specialist Handbook ch. 16, Oxford Handbook of ENT ch. 6, Dhingra ch. 8, 10 and K. J. Lee ch. 22. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

MIDDLE EAR & MASTOID · TRAUMA

Luxation of the Ossicular Chain

IMAGING FINDINGS

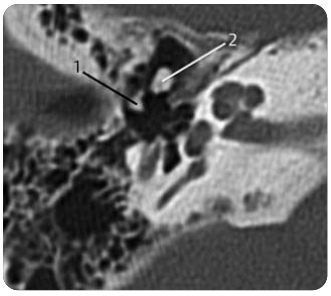
- Separation of the incudostapedial joint; incus rotated or displaced
- Loss of the "ice-cream cone" between malleus head and incus body
- Associated temporal bone fracture through the canal and middle ear

TOP DIFFERENTIALS

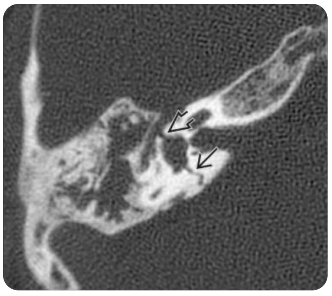
- Ossicular erosion from chronic otitis
- Congenital ossicular malformation
- Partial volume on a single slice

CHECKLIST

- Compare with the other ear on the same slice
- Check facial canal, tegmen and otic capsule



CT, axial. Traumatic luxation with dislocation of the connection between incus (1) and malleus (2) is seen. The typical icecone configuration of the…



Axial bone CT in a 14-year-old following severe head injury shows a medial transverse, otic capsule-violating fracture traversing the vestibule, with…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 968.